SECTION A: Short Answer (1–20)

1. Define AR, VR, and MR.
2. What is Reality–Virtuality Continuum?
3. Define immersion and presence.
4. List types of VR systems.
5. What is marker-based AR?
6. Define AR pipeline.
7. What is field of view (FOV)?
8. Define latency in VR.
9. What is graphics pipeline?
10. Define rasterization.
11. What is stereoscopic vision?
12. Define binocular disparity.
13. What is a scene graph?
14. Define Level of Detail (LOD).
15. What is culling?
16. Define tessellation.
17. What is animation?
18. Define haptic feedback.
19. What is force feedback?
20. What is OpenXR?

🧠 SECTION B: Medium Questions (21–40)

21. Explain AR vs VR vs MR.
22. Describe VR system components.
23. Explain AR tracking techniques.
24. Describe graphics pipeline stages.
25. Compare rasterization and ray tracing.
26. Explain stereoscopic display principles.
27. Describe VR hardware systems.
28. Explain scene graph and hierarchy.
29. Describe coordinate systems in VR.
30. Explain transformations (translation, rotation, scaling).
31. Describe bounding volumes.
32. Explain tessellation and its role.
33. Explain LOD techniques.
34. Compare culling methods.
35. Explain lighting models in VR.
36. Describe animation techniques.
37. Explain behavioral modeling.

38. Describe physics engines.
39. Explain collision detection.
40. Explain haptic feedback systems.

🧠 SECTION C: Long Answer (41–50)

41. Explain AR/VR system architecture.
42. Describe rendering pipeline in VR systems.
43. Explain stereoscopic vision and displays.
44. Discuss scene graphs and transformations.
45. Explain LOD and culling techniques in detail.
46. Discuss animation and behavioral modeling.
47. Explain physics engines and collision response.
48. Describe haptic interaction and force feedback systems.
49. Explain Unity architecture and workflow.
50. Discuss AR/VR applications in industry, medical, and education.

🔢 SECTION D: NUMERICAL QUESTIONS

🔹 Physics & Simulation

51. A virtual object of mass **5 kg** is accelerated at **2 m/s²**. Find the force applied.
52. A VR ball has mass **3 kg**. Calculate the gravitational force acting on it.
53. An object starts from rest and accelerates at **4 m/s²** for **3 seconds**. Find its final velocity.
54. A virtual object of mass **2 kg** moves with velocity **6 m/s**. Find its momentum.
55. A virtual spring has stiffness **k = 250 N/m**. If displacement is **0.02 m**, calculate the force applied.
56. For a spring system, stiffness **k = 300 N/m** and displacement **0.01 m**. Find the force generated.
57. A spring is compressed by **0.03 m** with stiffness **k = 200 N/m**. Calculate energy stored.

🔹 Collision & Interaction

58. A ball hits the ground with velocity **12 m/s** and rebounds at **8 m/s**. Find the coefficient of restitution.
59. Two objects have radii **2 units** and **4 units**, and distance between centers is **5 units**. Determine whether collision occurs.
60. A VR scene has **600 objects**, out of which **200 are culled**. How many objects are rendered?
61. Near plane = **1**, far plane = **15**. An object is located at **z = 18**. Determine whether it will be rendered.

🔹 Lighting Numericals

62. Ambient light intensity Ia=0.5I_a = 0.5Ia=0.5, reflection coefficient ka=0.6k_a = 0.6ka=0.6. Find final intensity.
63. Light intensity Il=1.0I_l = 1.0Il=1.0, diffuse coefficient kd=0.7k_d = 0.7kd=0.7, angle θ=60°θ = 60°θ=60°. Calculate diffuse lighting intensity.
64. Given Il=1.0I_l = 1.0Il=1.0, kd=0.8k_d = 0.8kd=0.8, angle θ=0°θ = 0°θ=0°. Find intensity.
65. For specular reflection: Il=1.0I_l = 1.0Il=1.0, ks=0.5k_s = 0.5ks=0.5, angle θ=45°θ = 45°θ=45°, shininess factor n=2n = 2n=2. Find intensity.
66. Using Phong model:
    Ia=0.3,ka=0.5I_a = 0.3, k_a = 0.5Ia=0.3,ka=0.5,
    Il=1.0,kd=0.6,θ=60°I_l = 1.0, k_d = 0.6, θ = 60°Il=1.0,kd=0.6,θ=60°,
    ks=0.4,α=30°,n=2k_s = 0.4, α = 30°, n = 2ks=0.4,α=30°,n=2.
    Find total intensity.
67. Two light sources contribute intensities **0.4** and **0.5**. Find total intensity.

🔹 Performance & Optimization

68. A VR system renders **1200 objects**, after LOD optimization **400 objects are simplified**. How many remain high-detail?
69. A scene originally runs at **60 FPS**, after optimization improves to **90 FPS**. Calculate percentage improvement.
70. A VR system updates haptic feedback at **500 Hz**. Calculate time per update.